

LINKED LIST

PROBLEM 6

QUESTION:

DS	Session	Linked List	Question Information	Level 1 Challenge 36
Question description A long time ago, there was a desolate village in India. The ancient buildings, streets, and businesses were deserted. The windows were open, and the stairwell was in disarray. You can be sure that it will be a fantastic area for mice to romp about in! People in the community have now chosen to provide high-quality education to young people in order to further the village's growth. As a result, they established a programming language coaching centre. People from the coaching centre are presently performing a test. Create a programme for the GetNth() function, which accepts a linked list and an integer index and returns the data value contained in the node at that index position. Example Input: 1->10->30->14, index = 2 Output: 30 The node at index 2 is 30 Constraints 1 < N < 1000 1 < X < 1000 1 < I < 1000 Input Format First line contains the number of datas- N. Second line contains N integers(the given linked list). Third Line Index I				

Output Format
First Line indicates the linked list
second line indicates the node at indexing position

Logical Test Cases

Test Case 1 INPUT (STDIN) 4 1 2 3 4 2 EXPECTED OUTPUT Linked list:-->4-->3-->2-->1 Node at index=2:3	Test Case 2 INPUT (STDIN) 4 11 21 31 41 3 EXPECTED OUTPUT Linked list:-->41-->31-->21-->11 Node at index=3:21
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Mandatory Test Cases

Test Case 1 KEYWORD struct node	Test Case 2 KEYWORD struct node *next;	Test Case 3 KEYWORD int GetNth(struct node* head,int index)
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Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
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CYCLOMATIC COMPLEXITY

10

TOKEN COUNT

385

NLOC

74

SOURCE CODE:

```
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

int GetNth(struct node* head, int index) {
    struct node* current = head;
    int count = 1;
    while (current != NULL) {
        if (count == index) {
            return current->data;
        }
        count++;
        current = current->next;
    }
    return -1;
}

int main() {
    int n, val, index;
    if (scanf("%d", &n) != 1) return 0;
    struct node* head = NULL;
    for (int i = 0; i < n; i++) {
        scanf("%d", &val);
        struct node* new_node = (struct node*)malloc(sizeof(struct node));
        new_node->data = val;
        new_node->next = head;
        head = new_node;
    }
    scanf("%d", &index);

    printf("Linked list:");
    struct node* temp = head;
    while (temp != NULL) {
        printf("-->%d", temp->data);
        temp = temp->next;
    }
    printf("\n");

    int ans = GetNth(head, index);
    printf("Node at index=%d:%d\n", index, ans);
    return 0;
}
```

OUTPUT:

```
4
1 2 3 4
2
Linked list:-->4-->3-->2-->1
Node at index=2:3
```

```
4
11 21 31 41
3
Linked list:-->41-->31-->21-->11
Node at index=3:21
```